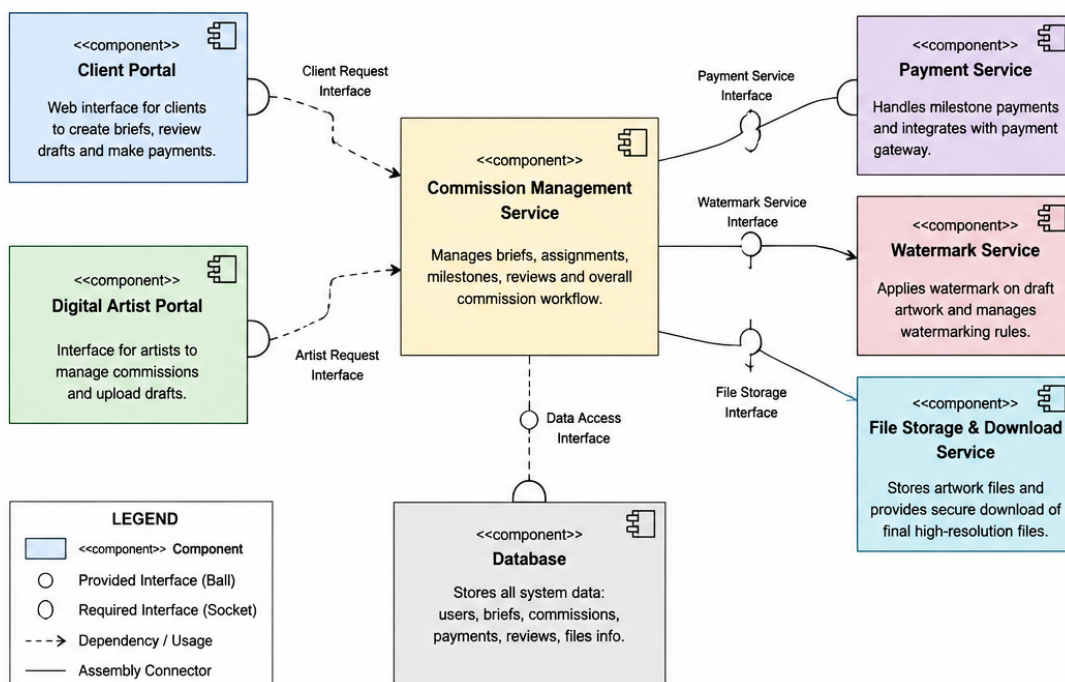


LAB 2 – COMPONENT DIAGRAM

Digital Art Commission & Watermarking Portal



ARCHITECTURE JUSTIFICATION

Selected Architecture:

Layered Architecture

1. Reasons for choosing Layered Architecture

- Separation of Concerns:** The system has a clear separation between user interfaces (Client Portal, Artist Portal), business logic (Commission Management Service), and services (Payment, Watermark, File Storage, Database). This makes the system modular and easier to maintain.
- Scalability and Maintainability:** Each layer can be developed, tested and updated independently. For example, the Payment Service or Watermark Service can be scaled or modified without affecting the user interface.

2. Security Advantage

- Sensitive operations like payments and file downloads are handled by dedicated services, keeping them isolated from the presentation layer. This reduces the risk of exposing sensitive data and improves access control.

3. Performance Advantage

- Services like Watermark Service and File Storage are separated, allowing independent scaling and caching strategies. This improves response time for draft uploads/downloads and overall system performance.

COMPONENTS USED (5+ Components)

- Client Portal
- Digital Artist Portal
- Commission Management Service
- Payment Service
- Watermark Service
- File Storage & Download Service
- Database

INTERFACES USED (4+ Interfaces)

- Client Request Interface
- Artist Request Interface
- Payment Service Interface
- Watermark Service Interface
- File Storage Interface
- Data Access Interface

COMPONENT DIAGRAM DELIVERABLE

- This diagram shows components, provided interfaces (ball), required interfaces (socket) and dependencies between them.

OUTPUT FORMAT

- PDF or PNG